Raghul Krishnan

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 4+ years of enterprise experience building agentic AI systems, API platforms, and developer tooling for a Fortune 500 HR technology company processing billions in annual transactions across 800,000+ clients. Architected a multi-agent autonomous development system and co-created an internal AI platform now used daily across engineering teams, earning company-wide Silver-level recognition. Trusted to lead ambiguous initiatives from technical discovery through production delivery, developer enablement, and cross-team adoption. Published co-author of research on auditable AI content licensing and safe agent autonomy.

PUBLICATIONS

Aegon: Auditable AI Content Access with Ledger-Bound Tokens and Hardware-Attested Mobile Receipts
arXiv:2604.06693, Apr 2026. Tamper-evident AI content licensing protocol pairing JWT licensing claims with a Merkle-tree audit ledger, with hardware-attested compliance receipts for on-device Android AI agents.

Mini Me: Personal AI That Was There. Continuum, a Context Substrate for Measurable Presence and Safe Action
Qeios, doi:10.32388/626YU5, Aug 2026. Architecture for personal AI assistants that keep context across a person's devices and conversations, and act on their behalf only through explicit, revocable permissions set per area of life.

PROFESSIONAL EXPERIENCE

Senior Software Engineer | **Mindex Technologies (Paychex)** *Jan 2022 - Present · Remote*

Promoted to Senior Software Engineer, 2025

AI Platform & Agentic Developer Tooling *2025 - 2026*

- **Architected Story-Dev, a multi-agent autonomous development system** that takes a ticket to PR-ready code through a 9-phase lifecycle with 7 specialist agents, deterministic safety guards, and human approval gates. Piloted on the team and validated by autonomously driving production stories end to end, including resolving PR review comments.
- **Co-created a custom AI developer platform (MCP server)** integrating 90+ tools across 8 enterprise systems into a unified natural-language interface, letting engineers investigate code, logs, deployments, and delivery workflows directly from the IDE. Now in daily use across multiple teams, cutting technical research and documentation effort from 5-6 days to 2-3 days.
- **Launched an internal AI plugin marketplace** packaging 5 reusable agents and skills as one-command installs for story delivery, PR review, technical research, engineering standards, and repository knowledge, establishing the program's standard for sharing AI capability.
- **Designed an autonomous incident remediation system** that correlates logs, container health, deployment history, and recent code changes to identify root cause and raise the fix as a pull request. Validated on a live production incident and projected to cut resolution time from 2-6 hours to under 30 minutes.
- **Scaled AI adoption across a 300+ person engineering organization** through 6+ cross-organizational presentations, hands-on onboarding, and reusable enablement documentation. Earned **company-wide Silver-level recognition** for process automation and program-wide knowledge sharing.
- **Authored the IPA Engineering Confidence System proposal**, defining a 90-day Forward Deployed Engineer pilot that links code intent, test coverage, release evidence, and incident learning into a single loop, targeting 40% less release-evidence effort and 30% faster failure triage.

Partner Integration Platform *2025 - 2026*

- **Designed end-to-end partner onboarding orchestration** consolidating 30+ manual API calls across multiple services into a single resumable, idempotent endpoint, reducing partner setup time from weeks to hours.
- **Architected a database-driven UI component system** enabling new partners to launch with full user-facing UI purely from configuration, eliminating frontend code changes entirely.
- **Delivered end-to-end provisioning for two partner product launches**, authoring the test plans that became the template for later rollouts. Cited by the API program's lead architect as “some of the best test cases I’ve ever seen.”

- **Led 3 technical investigations** on integration patterns, eligibility rules, and data cutover approaches that shaped architecture decisions adopted across the broader engineering program.

Core Platform & Mobile Banking *2022 - 2024*

- **Key contributor to the company's flagship mobile banking app** from inception through launch on Google Play and Apple App Store with zero critical defects across all releases. Recognized as **Product of the Year (2024)**.
- **Built an in-house performance monitoring system** for frontend load times, eliminating a $20,000+/year third-party dependency while cutting time-to-detect critical issues by 50%.
- **Improved CI/CD pipeline efficiency by 25%+** for mobile builds, accelerating deployment cycles and shortening developer feedback loops across the team.
- **Owned the authentication integration** after a team member's departure, leading the migration of login flows across all environments with zero downtime.
- **Designed scalable notification and webhook systems** for event-driven communications, building extensible architecture patterns reused by 4+ subsequent partner integrations.

Technical Leadership

- **Stepped into senior technical leadership** during a critical team transition, becoming the solution architect's go-to partner for architecture decisions, sprint planning, and cross-team coordination. Consistently described as the first to speak up and challenge designs in planning sessions.
- **Mentored junior and offshore engineers** through their first full-stack features, authored onboarding documentation, and bridged communication gaps with international team members.
- **Led complex data migrations** for employee onboarding workflows, coordinating across 3+ service teams while ensuring data integrity for 20+ operational teams. Recognized by the API program's lead architect for a “stellar” migration and test plan.
- **Received an “Exceeding” performance rating**, recognized as “a cornerstone for the team, continuously demonstrating technical skill and strong commitment to the team's success.”

Software Developer Co-op | **Solü Technology Partners** *Sep 2019 - May 2020 · Rochester, NY*

- Built a Spring Boot and AWS RDS backend at 95%+ test coverage, an Angular analytics portal, and Jenkins pipelines.

Software Developer Co-op | **CHAMPS Software** *May 2020 - Aug 2020 · Crystal River, FL*

- Built a cross-platform React Native app for Web, Android, and iOS with offline sync and real-time chat.

Software Developer | **King's College London** *Mar 2021 - Dec 2021 · Rochester, NY*

- Built a React/Next.js frontend capturing patient progress data for a healthcare research program.

TECHNICAL SKILLS

Languages	Java, JavaScript/TypeScript, Kotlin, Go, Python, SQL, Bash
AI & Agentic Systems	Multi-Agent Orchestration, Agentic Workflows, LangGraph, Model Context Protocol (MCP), RAG, Guardrails, LLM Integration & Model Routing, Human-in-the-Loop Systems
Backend	Spring Boot, Spring Data JPA, Kafka, REST APIs, Microservices, OAuth 2.0 / OIDC, JWT
Frontend	Angular, React, React Native, Next.js, Expo, HTML/CSS, Android Native
Cloud & Infrastructure	Kubernetes / OpenShift, AWS (EC2, RDS, S3, DynamoDB), Docker, Jenkins
Data	Oracle, MySQL, MongoDB, SQL Server
Tools & Practices	Git, Splunk, Gradle, Artifactory, Agile/Scrum, JUnit, Mockito, Postman, Jira, Confluence

EDUCATION

University of the Cumberlands | M.S. in Computer Science *GPA: 3.8 / 4.0 · Dec 2025*

Rochester Institute of Technology | B.S. in Computing and Information Technologies *GPA: 3.9 / 4.0 · Dec 2021*

PROFESSIONAL DEVELOPMENT

Forward Deployed Engineering Program | FDE Academy *Aug 2026 - Present*

36-week program in client discovery and production agentic delivery: LangGraph, MCP, enterprise RAG, guardrails, AWS AgentCore, multi-agent systems, and SLM fine-tuning.